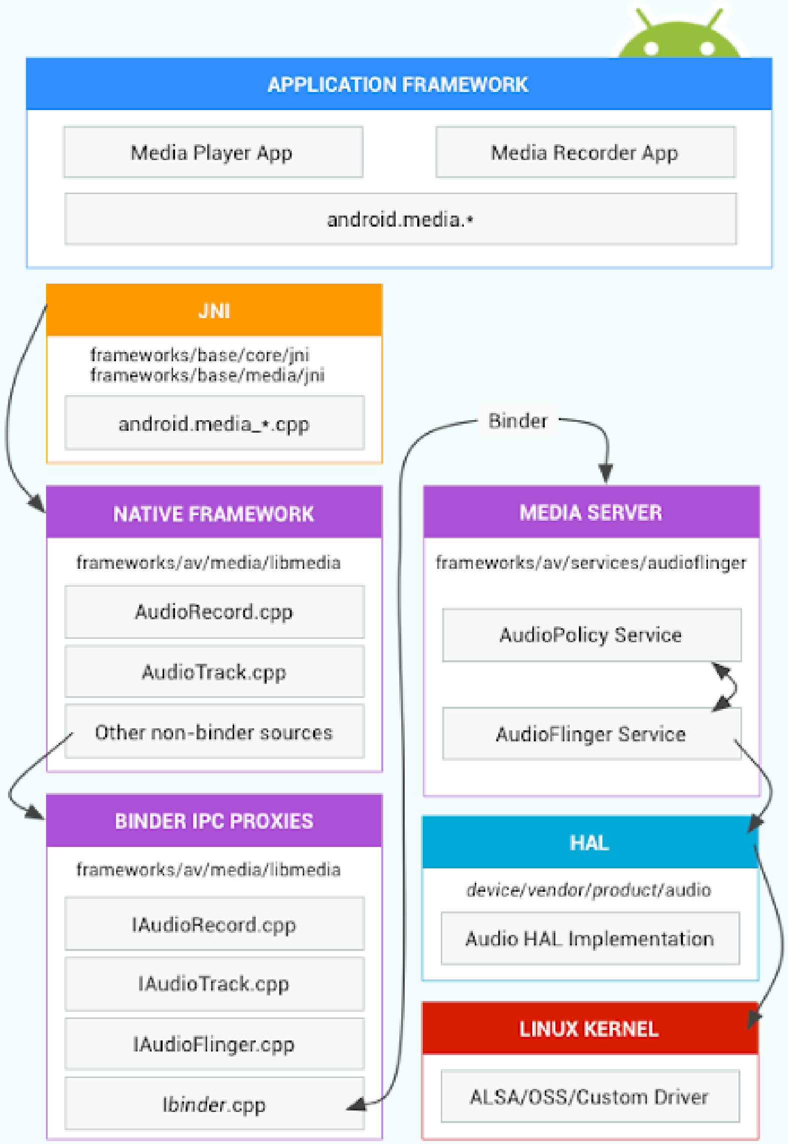
Android Audio Tutorial [Part Two] : Android Audio Architecture

- May 10, 2018

AudioArchitecture.

Android audio architecture defines how audio functionality is implemented and points to the relevant source code involved in the implementation.



Let us look at some of these components in the above diagram and try and see what are they needed for.

Application framework

The application framework includes the app code, which uses the android.media APIs to interact with audio hardware. Internally, this code calls corresponding JNI glue classes to access the native code that interacts with audio hardware.

JNI

The JNI code associated with android.media calls lower level native code to access audio hardware. JNI is located in frameworks/base/core/jni/ and frameworks/base/media/jni.

Native framework

The native framework provides a native equivalent to the android.media package, calling Binder IPC proxies to access the audio-specific services of the media server. Native framework code is located in frameworks/av/media/libmedia.

Binder IPC

Binder IPC proxies facilitate communication over process boundaries. Proxies are located in frameworks/av/media/libmedia and begin with the letter "I".

Media server

The media server contains audio services, which are the actual code that interacts with your HAL implementations. The media server is located in frameworks/av/services/audioflinger.

HAL

The HAL defines the standard interface that audio services call into and that you must implement for your audio hardware to function correctly. The audio HAL interfaces are located in hardware/libhardware/include/hardware. For details, see audio.h.

Kernel driver

The audio driver interacts with your hardware and HAL implementation. You can use Advanced Linux Sound Architecture (ALSA), Open Sound System (OSS), or a custom driver (HAL is driver-agnostic).

Android Audio architecture

Unknown *4 February 2019 at 21:39*

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REPLY

shaan watson *17 February 2021 at 22:19*

school auditorium sound system, I have read all the comments and suggestions posted by the visitors for this article are very fine,We will wait for your next article so only.Thanks!

REPLY

Ganesh Kumar *20 September 2021 at 04:18*

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Ossaward *15 November 2021 at 02:06*

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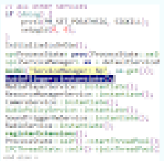
Android Audio Tutorial [Part Three] : AudioFlinger Introduction and Initialization

- May 11, 2018



AudioFlinger We know that AudioFlinger (Sometimes called AF) is the core of the entire audio system in Android. It is a system service which starts during boot and enables the platform for audio related use-cases in the following ways. 1.   Provide the access interface for the upper layer for using Audio. 2. …

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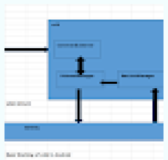
- May 22, 2018



Introduction. When we insert an sdcard/usb drive in an Android device, The system automatically mounts them for use. The vold [short for Volume Daemon] is the service responsible for detecting and mounting/unmounting of all the external or extended storage media in Android like usb, sdcard, cdrom …

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Android Audio Tutorial [Part One] : Introduction

- May 09, 2018

Sound Card. A sound card (also known as an audio card) is an internal expansion card that provides input and output of audio signals to and from a computer under control of computer programs. The term sound card is also applied to external audio interfaces used for professional audio applications. …

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